COMPSAC査読分析

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COMPSAC 査読結果

Review1

——— Originality ——— SCORE: 2 (Accept) ——— – Quality ——— SCORE: 2 (Accept) ——— Relevance ——— SCORE: 3 (Strong Accept) ——— Presentation ——— SCORE: 2 (Accept) ——— Overall Recommendation ——— SCORE: 2 (Accept)

Strong Aspects: The paper presents a well-structured approach to improving software system reliability using an automated framework that combines recent machine learning techniques with system-level monitoring and analysis.

The suggested methodology is a practical pipeline that collects runtime data, analyzes system behavior, and employs data-driven techniques to discover performance and reliability issues.

The experimental evaluation indicates measurable advances in system behavior analysis, as well as the suggested framework's usefulness across sample scenarios.

Overall, the paper tackles an important issue in modern software systems and proposes a workable solution that may be implemented in real-world scenarios.

Weak Aspects: The evaluation is performed on a limited number of scenarios and system configurations.

While the results show that the proposed method is effective, more experiments with larger datasets and more different system contexts would help to evaluate the framework's robustness and wide applicability.

現在の実験の目的提案手法である DaTopLinks の有効性を検証するため。

これは、何の有効性か分からないのがよくない。実環境での有効性と捉えられてもおかしくない。

修正版 実験の目的

- 早期確定機能により不要な測定を削減できること
- 多宛先のリンクを同時に考慮することでどの程度測定コスト削減できるか
- 代表的な量子ノイズ環境において、規定の信頼性レベルを達成できることはできるか

を検証するために実験した
と書き替える。
これは、研究の問いの

- 各宛先内での最適リンクの特定と、宛先間の Top-K 選択を同時に（結合して）行うことで、不要な測定を削減する適応型アルゴリズムを設計できるか？
- 提案手法は、代表的な量子ノイズ環境において、規定の信頼性レベルを達成できるか？

に対応している。
現在の実験では、

- 多宛先のリンクを同時に考慮することでどの程度測定コスト削減できるか

という問いに対する回答が分からない。
比較対象の手法が、

- DaTopLinks-NEC

早期確定をしない手法

- Fidelity-K

重要度、多宛先を考慮しない手法

であり、多宛先のみを考慮しない手法が含まれていないからである。Fidelity-K に関しては、多宛先を考慮しないし、重要度も考慮しないため、測定コストが増える原因が、多宛先を考慮しないことが原因なのか、重要度も考慮しないことが原因なのか結果からは分析できない。よって多宛先のみを考慮しない手法 (linkselfie を各宛先ごとに実行し、全ての宛先で 1 つのリンクを決定した後 topK 選択を行なう手法) を比較手法で追加するべきだと考えた。←現在実装中です

Recommended Changes: The paper could be improved by extending the evaluation to cover more datasets and system configurations.

Further comparisons with other current approaches to automated system monitoring and analysis would help to highlight the suggested method's advantages.

Additional discussion on scalability and deployment considerations in large-scale environments will greatly improve the contribution.

Review2

——— Originality ——— SCORE: 2 (Accept) ——— – Quality ——— SCORE: 2 (Accept) ——— Relevance ——— SCORE: 2 (Accept) ——— Presentation ——— SCORE: 2 (Accept) ——— Overall Recommendation ——— SCORE: 2 (Accept)

Strong Aspects

The paper addresses a relevant problem in quantum networks by incorporating destination importance into high-fidelity link selection.

It also presents a clear algorithmic design with theoretical analysis and simulation results across multiple noise models.

Weak Aspects

The novelty is moderate, as the work mainly extends existing link-selection and bandit-based approaches to a demand-aware Top-K setting.

The evaluation is limited to simulations, and performance under amplitude damping shows reduced robustness due to model mismatch.

Recommended Changes

The paper would benefit from a clearer explanation of its novelty relative to prior methods such as LinkSelFiE and related Top-K selection work.

It would also be helpful to expand the discussion of robustness under non-Pauli noise and practical deployment considerations.

REVIEW3

———— Originality ———— SCORE: 1 (Weak Accept) ——— Quality ———— SCORE: 1 (Weak Accept) ———— Relevance ———— SCORE: 2 (Accept) ———— Presentation ———— SCORE: 1 (Weak Accept) ———— Overall Recommendation ———— SCORE: 1 (Weak Accept)

Strong Aspects

The paper addresses an important problem in quantum networks by incorporating demand-aware link selection under limited resources.

The proposed DaTopLinks algorithm effectively combines intra- and inter-destination optimization with a clear design.

The inclusion of theoretical guarantees and evaluation across multiple noise models strengthens the contribution.

Weak Aspects

The novelty is somewhat incremental, as it builds on existing bandit-based link selection methods.

The evaluation relies only on simulations, limiting practical validation.

Additionally, the method depends on specific noise assumptions, and some sections are dense.

Recommended Changes

The authors should improve clarity by simplifying the algorithm description and adding intuitive explanations.

Expanding evaluation to more realistic scenarios or noise models would strengthen the work.

A clearer discussion of limitations and broader comparisons would also help.

REVIEW4

———— Originality ———— SCORE: 2 (Accept) ———— Quality ———— SCORE: 2 (Accept) ———— Relevance ———— SCORE: 2 (Accept) ———— Presentation ———— SCORE: 2 (Accept) ———— Overall Recommendation ———— SCORE: 2 (Accept)

Strong Aspects

The authors address a realistic gap—existing work ignores heterogeneous demand across destinations in quantum networks.

This work extends best-arm identification to a demand-aware Top-K selection problem, which is both meaningful and non-trivial.

Weak Aspects

The approach relies on simplified assumptions (e.g., fixed importance, star topology, and idealized conditions), which may limit its applicability to real-world, dynamic quantum networks.

Recommended Changes

Incorporate more realistic network conditions—such as dynamic traffic patterns, multi-hop topologies, and uncertain or time-varying destination importance—to improve practical applicability.

Need to extend the utility function and validate the approach through real-world experiments or testbed deployments to demonstrate robustness beyond simulation.